KENDRIYA VIDYALAYA SANGATHAN, RAIPUR REGION
FIRST PRE-BOARD EXAM (2024-25)

SUBJECT - SCIENCE

Class : X

MM : 80 **Time : 3 hours**

General Instructions:

1. All questions would be compulsory. However, an internal choice of approximately 33% would be provided. 50% marks are to be allotted to competency-based questions.
2. Section A would have 16 simple/complex MCQs and 04 Assertion-Reasoning type questions carrying 1 mark each.
3. Section B would have 6 Short Answer (SA) type questions carrying 02 marks each.
4. Section C would have 7 Short Answer (SA) type questions carrying 03 marks each.
5. Section D would have 3 Long Answer (LA) type questions carrying 05 marks each.
6. Section E would have 3 source based/case based/passage based/integrated units of assessment (04 marks each) with sub-parts of the values of 1/2/3 marks.

Section-A

Question 1 to 16 are multiple choice questions. Only one of the choices is correct. Select and write the correct choice as well as the answer to these questions.

1. The colour of the solution observed after 30 minutes of placing zinc metal to copper sulphate solution is
   (a) Blue (b) Colourless
   (c) Dirty green (d) Reddish Brown
2. Observe the given reactions and read the statements given below-
   $\mathbf{2PbO(s) + C(s) \rightarrow 2Pb(s) + CO_2(g)}$
   (i) Lead oxide is getting oxidised
   (ii) Carbon is getting oxidised
   (iii) Carbon dioxide is getting reduced
   (iv) Lead is getting reduced
   Which of the statements about the reaction below are correct?
   (a) (i) and (ii) (b) (i) and (iii)
   (c) (ii) and (iii) (d) (ii) and (iv)
3. What happens when dilute hydrochloric acid is added to iron fillings?
   (a) Hydrogen gas and Iron chloride are produced.
   (b) Chlorine gas and Iron hydroxide are produced.
   (c) No reaction takes place.
   (d) Iron salt and water are produced.
4. A solution turns red litmus blue, its pH is likely to be
   (a) 1 (b) 4
   (c) 5 (d) 10

5. A sample of soil is mixed with water and allowed to settle. The clear supernatant solution turns the pH paper yellowish-orange. Which of the following would change the colour of this pH paper to greenish-blue?

(a) Lemon juice (b) Vinegar
(c) Common salt (d) Baking soda

6. Which of the following methods is suitable for preventing an iron frying pan from rusting?

(a) Applying grease (b) Applying paint
(c) Applying a coating of zinc (d) All of the above

7. An element "A" is soft and can be cut with a knife. This is very reactive to air and cannot be kept open in the air. It reacts vigorously with water. Identify the element from the following

(a) Mg (b) Na
(c) P (d) Ca

8. In which of the following groups of organisms, food material is broken down outside the body and absorbed?

(a) Amoeba (b) Bread mould
(c) Cuscuta (d) Green Plants

9. If salivary amylase is lacking in the saliva, which of the following events in the mouth cavity will be affected?

(a) Proteins breaking down into amino acids (b) Starch breaking down into sugars
(c) Fats breaking down into fatty acids and glycerol (d) Absorption of vitamins

10. Lack of oxygen in muscles often leads to cramps among cricketers. This results due to

(a) conversion of pyruvate to ethanol (b) conversion of pyruvate to glucose
(c) non-conversion of glucose to pyruvate (d) conversion of pyruvate to lactic acid

11. There was a dysfunction in the cerebellum of a patient. Which of the following activities will get disturbed in this patient as a result of this?

(a) Salivation (b) Hunger control
(c) Posture and balance (d) Regulation of blood pressure

12. Which of the following is not a part of the female reproductive system in human beings?

(a) Ovary (b) Uterus
(c) Vas deferens (d) Fallopian tube

13. A child is standing in front of a magic mirror. She finds the image of her head bigger, the middle portion of her body of the same size and that of the legs smaller. The following is the order of combinations for the magic mirror from the top.

(a) Plane, convex and concave (b) Convex, concave and plane
(c) Concave, plane and convex (d) Convex, plane and concave

14. A student sitting on the last bench can read the letters written on the blackboard but is not able to read the letters written in his textbook. Which of the following statements is correct?

(a) The near point of his eyes has receded away
(b) The near point of his eyes has come closer to him
(c) The far point of his eyes has come closer to him
(d) The far point of his eyes has receded away

15. In the given food chain, suppose the amount of energy at the fourth trophic level is 5 kJ, what will be the energy available at the producer level?

Grass → Grasshopper → Frog → Snake → Hawk

(a) 5 k J (b) 50 k J
(c) 500 k J (d) 5000 k J

16. If a grasshopper is eaten by a frog, then the energy transfer will be from
(a) producer to decomposer (b) producer to primary consumer
(c) primary consumer to secondary consumer (d) secondary consumer to primary consumer

Question No. 17 to 20 consist of two statements – Assertion (A) and Reason (R).
Answer these questions by selecting the appropriate option given below:
A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true

17. **Assertion (A)**: Decomposition of vegetable matter into compost is an example of exothermic reactions.
Reason (R): Exothermic reactions are those reactions in which heat is evolved.
18. **Assertion (A):** The offspring produced by sexual reproduction is likely to adjust better in environmental fluctuation.
Reason (R): During the fusion of gametes there is mixing of genetic material from two parents.
19. **Assertion (A):** A ray of light travelling from a rarer medium to a denser medium slows down and bends away from the normal. When it travels from a denser medium to a rarer medium, it speeds up and bends towards the normal.
Reason (R): The speed of light is higher in a rarer medium than a denser medium.
20. **Assertion (A):** The concentration of harmful chemicals is least in human beings.
Reason (R): Man is at the apex of the food chain.

Section-B

(Question No. 21 to 26 are very short answer questions)

21. A shiny brown coloured element ‘X’ on heating in the air becomes black in colour. Name the element ‘X’ and the black-coloured compound formed.
22. How are fats digested in our bodies? Where does this process take place?

23. How the power and focal length of a lens related? You are provided with two lenses of focal lengths 20 cm and 40 cm, respectively. Which lens will you use to obtain more convergent light?

OR

An object 5 cm is placed at a distance of 20 cm in front of a convex mirror of radius of curvature 30 cm. Find the position, nature and size of the image.

24. In each of the following situations what happens to the rate of photosynthesis?
(a) Cloudy days
(b) No rainfall in the area
(c) Good manuring in the area
(d) Stomata get blocked due to dust

25. (a) Form a food chain consisting of the following organisms.

Insects, Hawk, Grass, Snake, Frog

(b) Which of these organisms will have the highest concentration of non-biodegradable chemicals? Name the phenomenon associated with it.

OR

What are trophic levels? Give an example of a food chain and state the different trophic levels in it.

26. Draw a schematic diagram of a circuit consisting of a battery of three cells of 2 V each, a 5 Ω resistor, an 8 Ω resistor, and a 12 Ω resistor, and a plug key, all connected in series. Calculate the net resistance through the circuit.

Section-C

(Question No. 27 to 33 are short answer questions)

27. You have been provided with three test tubes. One of them contains distilled water and the other two contain an acidic solution and a basic solution, respectively. If you are given only red litmus paper, how will you identify the contents of each test tube?

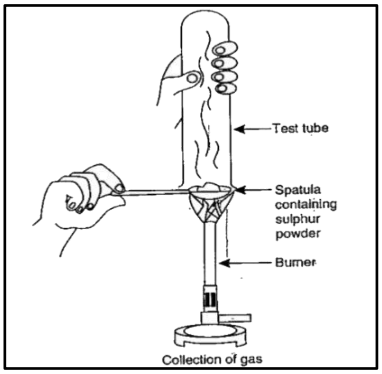
28. Samples of four metals A, B, C and D were taken and added to the following solution one by one. The results obtained have been tabulated as follows.

Metal	Iron (II) sulphate	Copper (II) sulphate	Zinc sulphate	Silver nitrate
A	No reaction	Reddish brown deposit	________	__________
B	Grey deposit	__________	No reaction	__________
C	No reaction	No reaction	No reaction	White shining deposit
D	No reaction	No reaction	No reaction	No reaction

Use the Table above to answer the following questions about metals A, B, C and D.
(i) Which is the most reactive metal?
(ii) What would you observe if B is added to a solution of Copper (II) sulphate?
(iii) Arrange the metals A, B, C and D in the order of decreasing reactivity.

OR

Pratyush took sulphur powder on a spatula and heated it. He collected the gas evolved by inverting a test tube over it, as shown in figure below.



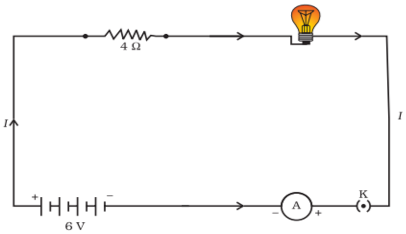
(a) What will be the action of gas on

(i) dry litmus paper?

(ii) moist litmus paper?

(b) Write a balanced chemical equation for the reaction taking place.

29. An electric lamp, whose resistance is 20 Ω, and a conductor of 4 Ω resistance are connected to a 6 V battery.



Calculate

(a) the total resistance of the circuit,

(b) the current through the circuit, and

(c) the potential difference across the electric lamp and conductor.

30. Draw excretory system in human beings and label the following organs of excretory system which perform following functions:

(i) form urine

(ii) is a long tube which collects urine from kidney

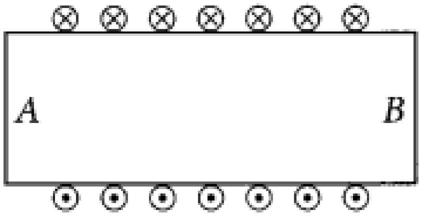
(iii) store urine until it is passed out.

31. “The sex of a newborn child is a matter of chance and none of the parents may be considered responsible for it”. Justify this statement with the help of a flow chart showing determination of sex of a newborn.

32. (a) Draw a ray diagram to show the refraction of light through a glass prism. Mark the angle of deviation on it.

(b) How will you use two identical glass prisms so that a narrow beam of white light incident on one prism emerges out of the second prism as white light? Draw and label the ray diagram.

33. Diagram shows the lengthwise section of a current carrying solenoid.



⊗ indicates current entering into the page, ⊙ indicates current emerging out of the page. Decide which end of the solenoid A or B, will behave as the north pole. Give reason for your answer. Also draw field lines inside the solenoid.

Section-D

(Question No. 34 to 36 are long answer questions.)

34. (a) Draw the structures for the following compounds.

(i) Ethanoic acid

(ii) Bromopentane

(iii) Butanone

(iv) Hexanal.

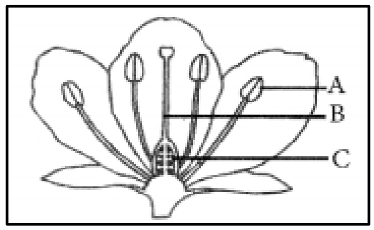
(b) Draw one structural isomers possible for bromopentane.

OR

(a) What is a homologous series? Explain with an example.

(b) How can ethanol and ethanoic acid be differentiated on the basis of their physical and chemical properties?

35. (a) Name the parts A, B and C shown in the following diagram and state one function of each.



(b) How does reproduction help in providing stability to populations of species?

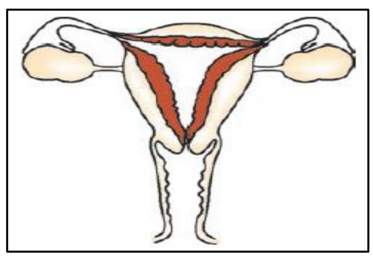
OR

(a) In the given figure, label the parts and mention their functions

(i) Production of egg

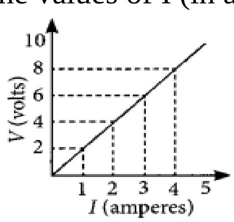
(ii) Site of fertilisation

(iii) Site of implantation



(b) Describe sexually transmitted diseases and mention the ways to prevent them.

36. (a) Study the V-I graph for a resistor as shown in the figure and prepare a table showing the values of I (in amperes) corresponding to four different values V (in volts).



How can we determine the resistance of the resistor from this graph?

Find the value of current for V = 10 volts.

(b) Calculate the resistance of a metal wire of length 2m and area of cross section 1.55×10^{-6} m², if the resistivity of the metal is 2.8×10^{-8} Ωm.

OR

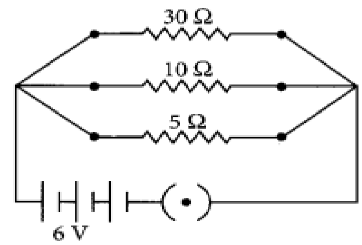
(a) Two wires A and B are of equal length and have equal resistances. If the resistivity of A is more than that of B, which wire is thicker and why?

(b) For the electric circuit given below calculate:

(i) current in each resistor

(ii) equivalent resistance of the circuit.

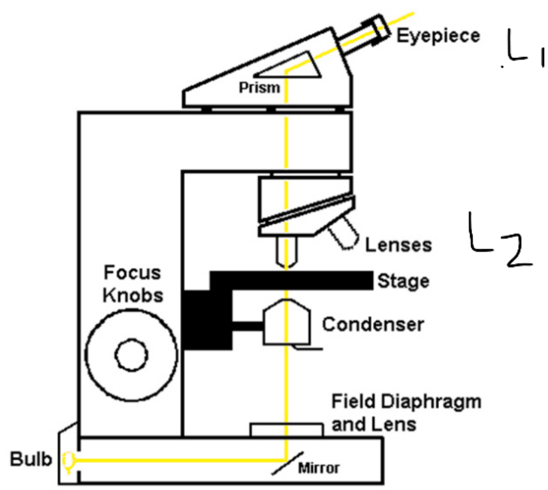
(iii) total current drawn from the battery, and



Section – E

(Question No. 37 to 39 are case-based/data -based questions.)

37. A compound microscope is an instrument which consists of two lenses L_1 and L_2. The lens L1, called objective, forms a real, inverted and magnified image of the given object. This serves as the object for the second lens L_2; the eye piece. The eye piece functions like a simple microscope or magnifier. It produces the final image, which is inverted with respect to the original object, enlarged and virtual.



Answer the following questions-

(a) What types of lenses must be L_1 and L_2? (1)

(b) What is the magnification (according to the new Cartesian sign convention) of the image formed by L_1? (1)

(c) If the power of the eyepiece (L_2) is 5 diopters and it forms an image at a distance of 80 cm from its optical centre, at what distance should the object be? (2)

38. Rohan needed a dilute acid during an experiment in the Science lab. He had conc. HCl only. He asked his teacher if he can make a dilute acid by adding water to it?

Teacher stopped him immediately from doing so and explained that the process of dissolving an acid or a base in water is a highly exothermic one. Care must be taken while mixing concentrated nitric acid or sulphuric acid with water. The acid must always be added slowly to water with constant stirring. If water is added to a concentrated acid, the heat generated may cause the mixture to splash out and cause burns. The glass container may also break due to excessive local heating. Look out for the warning sign (shown in Fig.) on the can of concentrated sulphuric acid and on the bottle of sodium hydroxide pellets.

Mixing an acid or base with water results in a decrease in the concentration of ions (H_3O^+/OH^-) per unit volume. Such a process is called dilution and the acid or the base is said to be diluted.

Answer the following questions for Rahul-

(a) How is the concentration of hydronium ions (H_3O^+) affected when a solution of an acid is diluted? (1)

(b) The process of dissolving an acid or a base in water is a highly exothermic one. Justify. (1)

(c) While diluting an acid, why is it recommended that the acid should be added to water and not water to the acid? (2)

39. Juhi was instructed to do a project as following-

- Fill a conical flask with water.
- Cover the neck of the flask with a wire mesh.
- Keep two or three freshly germinated bean seeds on the wire mesh.
- Take a cardboard box which is open from one side.
- Keep the flask in the box in such a manner that the open side of the box faces light coming from a window. (Given figure)
- After two or three days, you will notice that the shoots bend towards light and roots away from light.

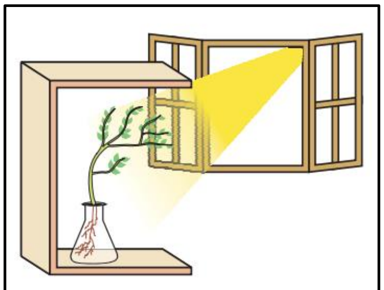


Figure 7.5
Response of the plant to the direction of light

Answer the following questions for Juhi-

(a) Now turn the flask so that the shoots are away from light and the roots towards light. Leave it undisturbed in this condition for a few days. Will the old parts of the shoot and root change direction? (1)

(b) Name the phenomenon responsible for growth of the shoot towards the light. (1)

(c) How do auxins promote the growth of the shoot towards the light? Explain with a suitable diagram. (2)

****